

An Investigative Study of Computer Vision Techniques for Red-Light Violation Detection

A Technical Seminar Report

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ABSTRACT

Traffic intersections are highly critical nodes within urban road networks. The violation of traffic signals—specifically red-light running (RLR)—is a primary cause of severe right-angle (T-bone) collisions, leading to significant fatalities and property damage on Indian roads and globally. Traditionally, red-light enforcement has relied on physical hardware such as inductive loop detectors buried beneath the asphalt. However, these systems are invasive, expensive to maintain, and prone to mechanical degradation. In recent years, automated computer vision-based object detection models have emerged as a scalable and highly efficient alternative.

This seminar report presents an investigative study of computer vision techniques specifically tailored for Red-Light Violation Detection (RLVD). The study dissects the dual-pipeline approach required for such systems: the simultaneous detection of the traffic light state and the tracking of vehicle trajectories. It explores the integration of classical image processing techniques, such as HSV color space thresholding for signal state recognition, alongside state-of-the-art deep learning architectures like YOLOv7 for high-speed vehicle detection and bounding box regression.

By analyzing the spatial relationship between detected vehicles and predefined virtual threshold lines (stop lines) during a "Red" signal state, intelligent systems can autonomously flag violators. This report provides a comprehensive comparative analysis of various detection algorithms, details the system architecture required for real-time video stream processing, and evaluates the operational challenges such as illumination changes and occlusion. The objective is to highlight how deep learning-based vision systems offer a robust, non-intrusive paradigm for intelligent traffic law enforcement.

INTRODUCTION

The exponential growth of vehicular traffic in developing nations has outpaced the scalability of manual traffic management. Amidst this congestion, the violation of traffic rules—particularly jumping red signals—has become a pervasive issue. A Red-Light Violation Detection (RLVD) system is an automated enforcement tool designed to capture instances where a vehicle crosses the stop line after the traffic signal has turned red.

1.1 Objectives

The primary objective of this seminar report is to systematically investigate the computer vision methodologies utilized to detect red-light jumping autonomously. The specific objectives include:

- To evaluate techniques for real-time Traffic Light Recognition (TLR), focusing on Region of Interest (ROI) extraction and color masking.
- To analyze advanced vehicle detection and tracking algorithms, specifically the application of YOLOv7 in complex traffic environments.
- To define the geometric and spatial logic required to determine when a tracked vehicle has breached a virtual stop line.
- To compare the efficacy of vision-based systems against traditional hardware sensor-based enforcement methods.

1.2 Scope of the Report

This report limits its scope to 2D vision-based systems using standard IP surveillance cameras. It covers the algorithmic workflow from raw frame extraction to violation classification. The investigation encompasses the mathematical principles of HSV color conversion, deep learning-based object detection, and threshold crossing logic. The findings are contextualized for deployment in Automated Traffic Monitoring Systems (ATMS) aimed at reducing manual policing overhead and improving intersection safety.

LITERATURE REVIEW

The enforcement of red-light compliance has undergone a significant technological evolution, moving from human observation to hardware sensors, and finally to intelligent video analytics.

2.1 Hardware-Based Sensor Systems

Historically, automated red-light cameras relied on inductive loop detectors or piezoelectric sensors installed beneath the road surface at the stop line. When a vehicle passed over the loop, it altered the magnetic field, triggering the camera if the traffic light was tied to a red phase. While highly accurate, these systems required destructive road work for installation and maintenance, causing severe traffic disruptions.

2.2 Early Computer Vision Approaches

To overcome the limitations of physical sensors, researchers turned to non-intrusive video image processing. Early computer vision systems relied on Background Subtraction algorithms (e.g., Gaussian Mixture Models) and frame differencing to detect moving blobs (vehicles). Virtual tripwires were drawn on the camera feed. If a foreground blob crossed the wire during a red phase, a violation was recorded. However, these methods suffered from "shadow interference," where the shadow of a vehicle in an adjacent lane could trigger a false positive. Furthermore, detecting the state of the traffic light often relied on direct hardwiring to the traffic controller rather than visual recognition.

2.3 Deep Learning and Modern Pipelines

The advent of Convolutional Neural Networks (CNNs) revolutionized RLVD systems. Instead of tracking mere moving pixels, systems could now recognize the semantic concept of a "car," "truck," or "motorcycle" using models like Faster R-CNN and YOLO. Modern literature focuses on a completely vision-based approach where both the vehicle and the traffic light are analyzed visually.

Recent studies utilizing the YOLOv7 architecture have demonstrated the ability to process dense traffic frames at over 60 frames per second (FPS) with high Mean Average Precision (mAP). By integrating YOLO-based vehicle tracking with traditional HSV color thresholding for the traffic signal ROI, modern systems achieve an end-to-end visual enforcement pipeline that requires zero physical connection to road sensors or signal controllers.

COMPUTER VISION TECHNIQUES FOR RED-LIGHT DETECTION

Detecting a red-light violation requires the synchronization of two parallel visual tasks: determining the current state of the traffic light and tracking the position of vehicles relative to the stop line.

3.1 Traffic Light State Recognition

Rather than hardwiring the camera to the traffic light controller, modern systems visually analyze the traffic light.

- **ROI Extraction:** A specific Region of Interest (ROI) encompassing the traffic light is manually defined or autonomously detected within the camera frame.
- **Color Space Conversion:** The ROI is converted from the standard BGR (Blue, Green, Red) color space to the HSV (Hue, Saturation, Value) color space. HSV is significantly more robust to variations in lighting conditions (e.g., cloudy vs. sunny days).
- **Thresholding and Masking:** Specific numerical ranges for Red, Yellow, and Green hues are defined. For instance, an `inRange` function creates a binary mask for each color.
- **Pixel Counting:** The algorithm counts the number of non-zero pixels in each mask. The state with the highest concentration of bright pixels determines the active signal state (e.g., if `red_pixels > green_pixels`, `signal_state = "Red"`).

3.2 Vehicle Detection and Bounding Box Regression

The system must identify vehicles approaching the intersection. YOLOv7 (You Only Look Once version 7) is utilized for this task due to its optimal balance of speed and accuracy.

- The neural network processes the frame and outputs bounding boxes (x, y, w, h) for every detected vehicle.
- The system calculates the centroid of the vehicle or, more accurately for spatial tracking, the bottom-center coordinate of the bounding box $(x + w/2, y + h)$, representing the vehicle's footprint on the road.

3.3 Virtual Stop Line and Violation Logic

The core of the RLVD system is the geometric logic that ties the two detection pipelines together.

- **Calibration:** A "Virtual Stop Line" is drawn on the video feed by assigning a specific Y-coordinate (or a line equation $y = mx + c$ for angled cameras) that matches the physical stop line on the road.
- **Condition Evaluation:** For every frame, the system evaluates a strict boolean condition:
 - IF `signal_state == "Red"`

- AND $\text{vehicle_bottom_y} > \text{virtual_stop_line_y}$ (Assuming the Y-axis increases downwards in image coordinates)
- THEN Flag as Signal Jump Violation.
- Data Capture: Upon flagging a violation, the system extracts the frame, crops the vehicle's bounding box, and logs the timestamp for evidentiary purposes.



Figure 3.1 COMPUTER VISION TECHNIQUES FOR RED-LIGHT DETECTION

SYSTEM ARCHITECTURE AND LOGIC PIPELINE

The architecture of an RLVD system is designed to process continuous RTSP video streams with minimal latency, ensuring that no fast-moving violator is missed between frames. This requires a robust, multi-threaded approach where frame capturing and processing run asynchronously.

4.1 Data Ingestion and Preprocessing Layer

The system receives the live video stream from intersection surveillance cameras via network protocols. The preprocessing layer extracts individual frames and resizes them to the input dimensions required by the YOLO network (e.g., 640x640) while maintaining a high-resolution copy in a buffer for evidence generation. To ensure real-time responsiveness without dropping frames, multi-threading techniques are heavily employed. Additionally, a Gaussian blur or similar noise reduction filter may be applied to mitigate artifacts caused by video compression or low-light conditions before passing the frame to the neural network.

4.2 The Dual-Processing Pipeline

The architecture branches into two concurrent processes:

- **Signal Analysis Branch:** Extracts the pre-defined traffic light ROI, applies the HSV masks, counts the active pixels, and outputs the string value ("Red", "Yellow", or "Green"). Adaptive thresholding is often utilized here to dynamically adjust the HSV ranges based on ambient light, preventing sunrise or sunset glare from causing false reads.
- **Vehicle Tracking Branch:** Passes the frame through the YOLOv7 weights. It extracts bounding boxes for classes such as cars, buses, and motorcycles, and updates their positional coordinates. Algorithms like DeepSORT or simple centroid tracking assign a unique integer ID to each vehicle.

4.3 Violation Analysis and Database Layer

The analytics engine acts as a logic gate. It continuously polls the outputs of the two branches. When the logic condition for a red-light jump is met, it packages the vehicle's bounding box data, the cropped image, the timestamp, and the signal state into a payload. To ensure legal validity, this payload is often cryptographically hashed or watermarked to prevent tampering. This payload is stored in a relational database (like MySQL) and utilizes RESTful APIs to communicate seamlessly with cloud servers. This triggers an alert to an admin dashboard for e-challan processing, allowing authorities to review the high-resolution evidence and automatically dispatch penalty notices.

COMPARATIVE ANALYSIS OF TECHNOLOGIES

To understand the shift toward deep learning-based visual systems, we compare various methodologies used in the industry for enforcement and detection.

Table 1: Comparison of Red-Light Enforcement Paradigms

This table compares the traditional hardware-based approach against the modern computer vision approach

Feature / Metric	Inductive Loop Detectors (Hardware)	Computer Vision (YOLOv7 + HSV Analysis)
Installation	Requires trenching and road cutting.	Non-intrusive; uses existing pole-mounted cameras.
Maintenance Cost	High (Loops break under heavy asphalt wear).	Low (Software-based, camera lens cleaning).
Detection Scope	Detects only the presence of a metallic mass.	Classifies vehicle type (car vs. bike) and tracks trajectory.
Two-Wheeler Detection	Poor (Low metallic footprint fails to trigger sensor).	Excellent (Visual detection via trained neural networks).
Signal Synchronization	Requires physical hardwiring to traffic controller.	Independent; visually reads the light state autonomously.

Table 2: Traffic Light State Recognition Techniques

Different visual methods exist for determining the color of the traffic light.

Technique	Methodology	Advantages	Disadvantages
Fixed ROI + BGR Thresholding	Checking pure RGB pixel values in a fixed box.	Extremely fast computation.	Fails completely under varying sunlight or shadows.
Fixed ROI + HSV Masking	Converting to Hue, Saturation, Value for color thresholding.	Highly robust to illumination changes and glare.	Requires camera to remain perfectly still (ROI shifts if camera moves).

Deep Learning (YOLO for Lights)	Training an object detector to find and classify the active light.	Immune to camera movement; highly accurate.	Computationally heavy; overkill if the camera is static.
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Table 3: Vehicle Detection Models for RLVD

The accuracy of the violation capture depends heavily on the bounding box precision at the stop line.

Detection Model	Architecture Type	Processing Speed	Suitability for High-Speed Intersection Logic
Background Subtraction (GMM)	Classical Pixel-based	Very High	Poor. Shadows trigger false stop-line crossings.
YOLOv3 / YOLOv4	Deep Learning (One-Stage)	Medium (30-45 FPS)	Good. Reliable but may miss highly occluded vehicles.
YOLOv7	Deep Learning (One-Stage)	Very High (60-160 FPS)	Excellent. Highest mAP allows for precise centroid calculation at the exact frame of crossing.

CHALLENGES AND LIMITATIONS

While investigative studies prove the superiority of YOLOv7 and HSV masking, real-world deployment faces several environmental and technical hurdles.

1. **Sunlight Glare and Illumination Halos:** During sunrise or sunset, direct sunlight hitting the camera lens can wash out the traffic light ROI, making it impossible for the HSV mask to distinguish between a red light and sun glare. Similarly, LED traffic lights can create "halos" that bleed into adjacent color channels.
2. **Night-time Occlusion and Headlight Washout:** At night, vehicle headlights can temporarily blind the camera sensor. The object detector may struggle to draw an accurate bounding box around the vehicle chassis, instead drawing it around the light beam, which prematurely triggers the virtual stop line.
3. **Heavy Vehicle Occlusion:** In dense Indian traffic, a large bus or truck can completely block the camera's line of sight to a smaller car or motorcycle crossing the stop line alongside it, leading to missed violations.
4. **PTZ Camera Drift:** If the system utilizes Pan-Tilt-Zoom (PTZ) cameras or if the mounting pole vibrates in heavy wind, the fixed coordinates of the virtual stop line and the traffic light ROI will drift, rendering the spatial logic inaccurate until manually recalibrated.

CAUSES & RISK FACTORS IN INTERSECTION SAFETY

► Main Causes of Red-Light Violations

- **The "Dilemma Zone":** Drivers approaching an intersection at high speed when the light turns yellow often face a split-second decision: brake hard or accelerate to beat the red. Misjudgement here is the leading cause of unintentional red-light running.
- **Distracted Driving:** Smartphone usage or in-cabin distractions cause drivers to entirely miss the visual cue of the changing traffic signal.

► Risk Factors Exacerbating Collisions

- **T-Bone Collision Dynamics:** Intersections force vehicles moving on perpendicular vectors to cross paths. A red-light runner typically impacts the side of another vehicle (T-bone). The sides of vehicles have the least crumple zone protection, resulting in exceptionally high fatality rates.

CONCLUSION & FUTURE SCOPE

Conclusion

This investigative study has demonstrated that Computer Vision techniques, particularly those leveraging the YOLOv7 deep learning architecture, provide a highly effective, non-intrusive solution for Red-Light Violation Detection. By abandoning legacy hardware like inductive loops, municipalities can deploy scalable systems that utilize existing CCTV infrastructure. The integration of a dual-pipeline architecture—utilizing HSV color masking for autonomous Traffic Light Recognition and bounding box regression for precise spatial tracking—creates a robust logic gate for flagging violators. The comparative analysis confirms that modern deep learning models overcome the traditional pitfalls of classic image processing, such as shadow interference, offering unprecedented accuracy and real-time processing capabilities. Ultimately, the deployment of such automated enforcement systems serves as a powerful deterrent, actively modifying driver behavior and securing complex urban intersections.

Future Scope

The future enhancement of RLVD systems lies in creating more autonomous, self-healing, and comprehensive algorithms.

- **Auto-Calibration Algorithms:** Future systems will utilize vanishing point detection and lane-line segmentation to autonomously redraw the virtual stop line and traffic light ROI if the camera moves or vibrates, eliminating the need for manual recalibration.
- **Trajectory Prediction for Pre-Emptive Alerts:** By tracking the velocity and deceleration rate of an approaching vehicle, future AI models could predict if a vehicle will run a red light before it actually crosses the line, potentially triggering a localized warning flash to cross-traffic.
- **Edge AI Deployment:** Transitioning the YOLOv7 inference from central servers directly to the camera hardware (Edge Computing) will reduce bandwidth dependency, allowing the system to process frames instantaneously and transmit only lightweight JSON violation data to the command center.

REFERENCES

1. Dr. S. Raj Anand, Dr. Naveen Kilari, Dr. D. Udaya Suriya Raj Kumar, "Traffic Signal Violation Detection using Artificial Intelligence and Deep Learning", International Journal Advanced Research Engineering and Technology (IJARET) of in Volume 12, Issue 2, February 2021 DOI: 10.34218/IJARET.12.2.2021.043
2. Wang, C. Y., Bochkovskiy, A., & Liao, H. Y. M. (2022). "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors." arXiv preprint arXiv:2207.02696.
3. Ji-hun Won, Dong-hyun Lee, Kyung-min Lee, Chi-ho Lin, "An Improved YOLOv3-based Neural Network for De-identification Technology", DOI: 10.1109/ICTC46691.2019.8939912
4. Abhiraj Biswas, Arka Prava Jana, Mohana, Sai Tejas S, "Classification of Objects in Video Records using Neural Network Framework", International Conference on Smart Systems and Inventive Technology (ICCSIT 2018) IEEE Xplore DOI: 10.1109/ICSSIT.2018.8748439
5. Fernández Llorca, D., et al. (2021). "Vision-based vehicle speed estimation: A survey." IET Intelligent Transport Systems.
6. Choudekar, P., Banerjee, S., & Muju, M. K. (2011). "Real time traffic light control using image processing." Indian Journal of Computer Science and Engineering, 2(1), 6-10.
7. Omachi, M., & Omachi, S. (2009). "Traffic light detection with color and edge information." In 2009 2nd IEEE International Conference on Computer Science and Information Technology (pp. 284-287).
8. Do, N. T., & Le, T. H. (2020). "Vehicle Speed Estimation in Surveillance Videos using Deep Learning and Camera Calibration." International Conference on Multimedia Analysis and Pattern Recognition (MAPR).
9. "YOLOv7: The Fastest Object Detection Algorithm (2022)" - viso.ai
10. GitHub - WongKinYiu/yolov7: Implementation of paper YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors DOI: 10.5281/zenodo.6825424